

CP page 10

#10 $y = f(x^2 + 3x - 5)$

$$\frac{dy}{dx} = f'(x^2 + 3x - 5) \cdot (2x + 3)$$

$$\left. \frac{dy}{dx} \right|_{x=1} = f'(1^2 + 3(1) - 5) (2(1) + 3)$$

$$\left. \frac{dy}{dx} \right|_{x=1} = f'(-1) (5)$$

$$\left. \frac{dy}{dx} \right|_{x=1} = 2 \cdot 5$$

$$\left. \frac{dy}{dx} \right|_{x=1} = 10$$

#11.

$$y = g(h(x))$$

$$y' = g'(h(x)) \cdot h'(x)$$

$$y' = g'(h(3)) \cdot h'(3)$$

$$y' = g'\left(\frac{9}{5}\right) \cdot \left(\frac{21}{25}\right)$$

$$y' = (-2) \left(\frac{21}{25}\right)$$

$$y' = \frac{-42}{25}$$

quotient
Rule.



$$* h'(x) = \frac{2x(x+2) - x^2}{(x+2)^2}$$

$$h'(x) = \frac{2x^2 + 4x - x^2}{(x+2)^2}$$

$$h'(x) = \frac{x^2 + 4x}{(x+2)^2}$$

$$h'(3) = \frac{21}{25}$$

$$* \text{ Also, } h(3) = \frac{9}{5}$$